

Integrated Information Without Its Causes or Consequences

recurrence $\neq \Phi$ on a real LLM, and a Φ -peak $\neq$ causal emergence at the neuromorphic edge — a two-front closed-negative

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Abstract

Two popular “consciousness-by-proxy” hypotheses claim that integrated information Φ is (i) *caused* by recurrence and (ii) a marker of genuine *emergence*. We test both on real substrates with one common Φ -proxy engine and pre-registered, deterministic falsifiers. **Front 1 (cause).** On a frozen Qwen2.5-1.5B, we inject a GRU adapter into the layer-14 residual stream and train it on a language-model objective (base frozen). Recurrence does *not* raise Φ above a parameter-matched feedforward (FF) control: the FF attains strictly higher Φ on every seed (REC band [2.49, 2.65] vs FF band [2.70, 2.82] bits, $\Delta\Phi = -0.166$), higher self-prediction, and lower perplexity — so capacity, not recurrence, carries the integration. **Front 2 (consequence).** On a leaky-integrate-and-fire ring swept order→edge→chaos, the edge-of-chaos Φ peak is micro-reducible: causal emergence $CE = EI_{\text{macro}} - EI_{\text{micro}}$ is negative at the edge under both an observational proxy ($CE = -0.877$, 3 coarse-grainings) and Hoel *interventional* effective information with a grouping search ($CE = -0.587$, 6 coarse-grainings) — nine coarse-grainings, all refuted. Integration thus dissociates from both its hypothesized cause and its hypothesized consequence. **Scope.** Φ - and CE-proxies are necessary-condition correlates, not phenomenal consciousness; we make no claim about subjective experience. All results recompute bit-for-bit from the companion/ ledger and the LAB/ verdict files.

1 Introduction

Integrated Information Theory (IIT) proposes Φ , the information generated by a system above and beyond its parts, as a measure tied to consciousness [1, 8, 11]. Two auxiliary claims travel with it in practice. First, a *causal* claim inherited from recurrent-processing accounts [6]: that feedback / recurrence is what *produces* high Φ , so adding recurrence to a feedforward system should raise it. Second, a *consequence* claim borrowed from causal-emergence theory [4, 5]: that a high- Φ regime — canonically the edge of chaos [2, 7] — is a site of genuine emergence, where a coarse macro description has *more* causal power than the micro.

We stress-test both on real substrates. A prior synthetic smoke (a rule-110 ring) reproduced the textbook recurrence-raises- Φ pattern; here we ask whether either claim survives contact with (a) a real trained transformer and (b) an interventional emergence measure. They do not.

Contributions.

1. A residual-stream LM-objective protocol (Front 1) showing that recurrence does *not* cause integrated information on a real LLM: a parameter-matched feedforward control attains strictly higher Φ , self-prediction, and lower perplexity, with non-overlapping seed bands.

2. A causal-emergence test (Front 2) — observational, then Hoel interventional with a grouping search — showing the edge-of-chaos Φ peak is micro-reducible across nine coarse-grainings.
3. Two dissociations: integration $\not\sim$ recurrence (cause) and integration $\neq$ emergence (consequence), each with deterministic recompute.

2 Background

The Φ -proxy used throughout is the whole-minus-best-bipartition effective information of a binary state series, a deterministic relative of IIT’s integration term [1, 8] and of transfer entropy [10]. Recurrence is realized by a GRU cell [3] contrasted against a parameter-matched feedforward stack; the host LLM is a decoder-only transformer [9, 12]. Causal emergence is computed in Hoel’s sense [4, 5]: effective information under a maximum-entropy (uniform) intervention, compared between the micro state space and a coarse-grained macro space.

3 Full Pipeline

Tier rubric. ■ closed-form identity · ■ numerical (reproducible script + persisted output) · ■ cited · ■ install/silicon-gated · ■ falsified (kept as honest negative).

#	Stage	Tier	Backing artifact
①	Φ /EI engine (whole—min-bipartition)	■	LAB-09/10 harness
②	Front 1: real-LLM residual finetune	■	verdict_phi_14.txt
③	Front 2a: observational causal emergence	■	verdict_emergence.txt
④	Front 2b: interventional EI + grouping search	■	verdict_emergence_interventional.txt
⑤	Cross-tier sanity (CAUSAL Φ bit-match)	■	§5
⑥	Live AKD1000 silicon emergence	■	LAB-10 pi5-akida (not run here)

Table 1: Pipeline stages. Stage 6 (live-silicon causal emergence) stays ■: it is the substrate-grounded follow, not claimed here.

4 Method

Φ -proxy (formula). For a binary state series S_t over K median-split channels,

$$\Phi = I(S_t; S_{t+1}) - \min_{A \sqcup B = S} [I(A_t; A_{t+1}) + I(B_t; B_{t+1})],$$

the whole effective information minus its best (minimum-information) bipartition; self_pred = $I(S_t; S_{t+1})$. The identical engine is used in both fronts (Front 2 sets S = spike vector).

Front 1 — does recurrence cause Φ ? Base Qwen2.5-1.5B (28 layers, hidden 1536), float32, frozen (requires_grad = False). A bottleneck adapter (width 16) is injected into the layer-14 residual stream by a forward hook, $h \leftarrow h + \text{Adapter}(h)$, LoRA-style with the up-projection zero-initialized so step 0 equals the frozen base. **REC** uses a GRUCell (recurrence over the token

axis); **FF** uses a parameter-matched 3-linear stack (52,336 vs 51,520 params, $\sim 1.6\%$ apart, favoring REC). Both are trained only on the next-token cross-entropy of an 8-sentence corpus (Adam, lr 0.01, 80 steps, seeds {42, 43, 44}); gradients flow through the frozen upper layers. We then measure Φ , self-prediction, and perplexity on the adapter-modified layer-14 residual. *Pre-registered falsifiers:* **F2L4** $\Phi_{\text{REC}} > \Phi_{\text{FF, band-max}}$; **F4** both adapters lower ppl vs base; **F5** $\text{self}_{\text{REC}} > \text{self}_{\text{FF}}$. Headline: supported = F2L4.

Front 2 — is the Φ -peak emergent? An $N=6$ leaky-integrate-and-fire ring is swept across regimes R1 (weak/silent), R2 (noise edge), R3 (tonic edge), R4 (over-driven), the same sweep that yields the Φ inverse-U on neuromorphic silicon (LAB-10). Causal emergence $\text{CE} = \text{EI}_{\text{macro}} - \text{EI}_{\text{micro}}$, with the micro state the 6 spike units and the macro a spatial coarse-graining (pairs or triples). *Tier 2a (observational):* $\text{EI} = I(S_t; S_{t+1})$ on natural trajectories, three coarse-grainings (OR/AND/XOR). *Tier 2b (interventional, Hoel):* the transition matrix is built by $\text{do}(s)$ — force each of the 2^6 states, one memoryless step, sample over noise — and EI is the mutual information under a *uniform* intervention; EI_{macro} is the *maximum* over six coarse-grainings (pairs OR/AND/XOR, triples majority/OR/AND). *Pre-registered falsifiers:* **F1** $\text{CE}(\text{edge}) > 0$; **F2** $\text{CE}(\text{edge}) > \text{CE}(\text{R1})$; **F3** $\text{CE}(\text{edge}) > \text{CE}(\text{R4})$. All thresholds fixed before the runs.

5 Verify

Cross-tier sanity. The Front 1 CAUSAL_frozen reference reproduces the independent LAB-09 stress-tier causal baseline to full precision — $\Phi = 2.8477363469573813$, $\text{self} = 3.9074609119958468$ — confirming the Φ engine, channel selection, corpus, and layer are byte-identical across the two harnesses. Every numeric below is emitted verbatim by the harness; the model judges nothing (deterministic plug-in mutual information).

6 Results

Front 1: recurrence does not cause Φ . Table 2. The parameter-matched feedforward control attains strictly higher Φ than the recurrent GRU on *every* seed (REC max 2.6531 < FF min 2.7047: non-overlapping bands), higher self-prediction, and lower perplexity. Both adapters learn the task (ppl 34.1 $\rightarrow$ ~ 1.1 – 1.6), so the comparison is fair. F2L4 and F5 fail; only F4 (both-learned) passes. The stress-tier’s lone positive — recurrence raising self-prediction — *reverses* here.

arm	Φ (bits)	self_pred	perplexity
CAUSAL_frozen (ref)	2.8477	3.9075	34.12
REC_lm (GRU)	2.5953 [2.49, 2.65]	3.7822	1.595
FF_lm (control)	2.7613 [2.70, 2.82]	3.9634	1.130

Table 2: Front 1, Qwen2.5-1.5B, layer 14, 3 seeds. $\Delta\Phi_{\text{REC-FF}} = -0.166$, $\Delta\text{self} = -0.181$. Bands non-overlapping in FF’s favor. Params: REC 52,336 vs FF 51,520. **F2L4 FAIL · F4 PASS · F5 FAIL.**

Front 2: the Φ -peak is micro-reducible. Table 3. The Φ inverse-U reproduces (R3 peak), but causal emergence is negative at the edge under both measurement families. Observationally, $\text{CE}(\text{R3}) = -0.877$ and is the *most* negative exactly where micro-information is richest; the sign is robust across OR/AND/XOR. Interventionally with a six-way grouping search, even the best macro recovers only 0.618/1.205 $\approx 51\%$ of micro EI ($\text{CE} = -0.587$). Nine coarse-grainings total: all refuted (F1–F3 fail).

regime	Φ	EI _{micro}	EI _{macro}	CE
<i>Tier 2a — observational (OR coarse-grain)</i>				
R1 weak	0.000	0.000	0.000	0.000
R2 edge	0.112	0.237	0.017	−0.220
R3 edge	0.844	0.925	0.047	− 0.877
R4 chaos	0.000	0.000	0.000	0.000
<i>Tier 2b — Hoel interventional, best of 6 groupings</i>				
R3 edge	—	1.205	0.618	− 0.587

Table 3: Front 2, $N=6$ LIF ring. The edge maximizes integration (EI_{micro}) but no macro out-informs micro. Robustness (CE at R3): OR −0.877, AND −0.850, XOR −0.916 (observational); best-of-6 −0.587 (interventional). **F1·F2·F3 FAIL** on both tiers.

The two dissociations. High integration coincides with neither its hypothesized cause nor its hypothesized consequence: Φ is not raised by recurrence (Front 1, $\text{FF} > \text{REC}$) and the Φ peak is not a site of macro causal emergence (Front 2, $\text{CE} < 0$). The edge of chaos is a richly integrated *micro* process.

7 Limitations and honest caveats

- **Memorization regime (Front 1).** ppl ~ 1.1 – 1.6 on 8 sentences means both adapters largely memorize; the REC-vs-FF Φ comparison is still fair (both overfit equally, FF still higher), but absolute values are corpus-specific. A held-out train/val split is the next tier.
- **Proxy, not full IIT.** Φ here is whole–min-bipartition over $K=6$ median-split channels, not IIT 4.0 big- Φ with purview / cause-effect structure search.
- **Memoryless closure (Front 2b).** The interventional TPM resets the membrane each step; with these regime parameters R1/R2/R4 become sub-threshold (EI = 0), so the decisive test is F1, which fails large. The leaky observational tier (where R2 is active) agrees in sign.
- **Grouping search is six hand-specified spatial coarse-grainings**, not the full $64 \rightarrow k$ partition space; a positive CE under some exotic partition is not strictly excluded, but the loss is large ($\sim$ half).
- **Sim, not silicon** (Front 2): the live AKD1000 emergence run stays ■.
- **Strong (causal) emergence, not phenomenal consciousness.** A negative says the signature is micro-reducible at this measure; it is silent on subjective experience, which these proxies cannot reach.

8 Reproducibility

Code and harnesses: <https://github.com/dancinlab/hexa-codex> under LAB/lab-09-consciousness-direct (Front 1, phi_llm_14_harness.hexa) and LAB/lab-12-causal-emergence/ (Front 2, causal_emergence_sim and causal_emergence_interventional_harness.hexa). Verbatim verdicts: verdict_phi_14.txt, verdict_emergence.txt, verdict_emergence_interventional.txt. Front 1 env: torch 2.12.0 + transformers 5.9.0, Apple MPS, ~ 152 s. Front 2: pure-hexa, \$0, ~ 1 – 3 s. Build: make (xelatex $\times 3$ + bibtex).

9 Conclusion

On real substrates, integrated information Φ comes apart from the two properties usually attributed to it: recurrence does not cause it (a capacity-matched feedforward integrates as well or better), and an edge-of-chaos Φ peak is not causal emergence (it is micro-reducible across nine coarse-grainings and an interventional measure). The detection of a Φ signature is therefore not, by itself, evidence of a recurrence mechanism or of “the whole exceeding its parts.” The single most important follow-up is the substrate-grounded tier: a live AKD1000 interventional-emergence run with a full coarse-graining search.

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